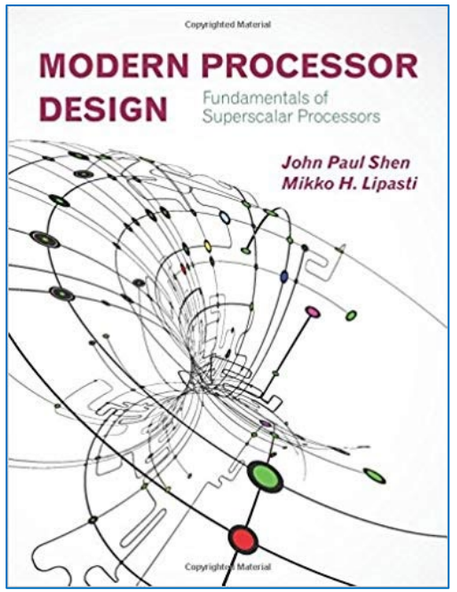
18-740: Course Overview (Fall 2026)

"Modern Computer Architecture & Design"

(Adventures in Processor Design and Computer Architecture)

John Paul Shen and Olivia Hsu

August 25, 2026



- **Complete pre-course survey**: https://forms.gle/ieRC3ZqH2N9DgFhZ7
- **Required Reading Assignment**:
  - **Chapters 1 & 2 of textbook:** *Modern Processor Design: Fundamentals of Superscalar Processors,* by John Shen and Mikko Lipasti, 2005; reissued by **Waveland Press Inc**, 2013.
- **Lab 0 should be done individually during Week 1.**
- **Recommended References**:
  - Gem5 Tutorial: https://www.youtube.com/watch?v=npbMXXoKltg
  - Learn Gem5: https://www.gem5.org/documentation/learning_gem5/introduction/

Electrical & Computer ENGINEERING

18-740: "Modern Computer Architecture and Design"

(Fall 2026)

Course Description: Computer architecture deals with the art and science of designing and integrating hardware subsystems, by spanning across the hardware/software interface to create a computer system that achieves functional correctness and meets design objectives in performance, power, energy, and cost. This course provides in-depth coverage of modern computer systems, with a special focus on System-on-Chips (SoCs) for mass-market mobile and edge computing devices, including smartphones, tablets, and diverse smart home/office devices. Topics covered span the design and evaluation of three major subsystems: (1) *Processing Subsystem:* superscalar out-of-order cores, multicore processors, asymmetric cores, and heterogeneous cores, AI/DL accelerators, (2) *Memory Subsystem:* main memory and DRAM, multicore cache hierarchy, and co-processor shared memory management, and (3) *Interconnect Subsystem:* Network on Chip (NoC), Multi-Die Packages, and Chiplets. Each subsystem will be studied from the perspective of several first-class design goals: performance, power consumption, and energy efficiency. This course places a strong emphasis on professional design tools (e.g., architecture/microarchitecture simulators, RTL design and synthesis tools, area/power/thermal modeling), and hands-on experimentation on physical hardware systems with the goal of preparing students to be highly competent **processor designers** and **computer architects** in the industry.

Recommended Pre/Co-requisites: 15-513/18-613; 18-447 (or a course based on *Computer Organization and Design* by Patterson & Hennessy). Having Verilog and Gem5 experience would be very helpful.

Required Textbook: *Modern Processor Design: Fundamentals of Superscalar Processors,* by John Shen and Mikko Lipasti, 2005; reissued by Waveland Press Inc, 2013. ISBN 10: 1-4786-0783-1, ISBN 13: 978-1-4786-0783-0.



Instructor: John Paul Shen
Office Location: CMU-SV, B23-114
Email Address: jpshen@andrew.cmu.edu
Office Hours on Zoom: Wed 6:00pm-7:00pm ET



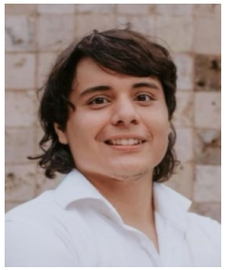
Co-Instructor: Olivia Hsu
Office Location: CIC 4121
EmaiAddress: owh@andrew.cmu.edu
Office Hours: Tuesdays/Thursdays 2:00-2:50pm ET



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Week	Date	Day	Class Activity	Labs
	Lab 0: Orientation (1 week)			Lab 0
Week 1	8/25	Tues	Lecture 1: Computer Architecture & Processor Design [**JS/OH**]	
	8/27	Thur	Lecture 2: Modern Superscalar Out-of-Order Processors [**JS**]	
	8/28	Fri	Recitation 1: Course Overview; Lab 0 and Lab 1 Assignments	
	Lab 1: CPU Design: Dynamic Branch Predictors			Lab 1
Week 2	9/1	Tues	Lecture 3: Advanced Dynamic Branch Prediction Techniques [**JS**]	
	9/3	Thur	Lecture 4: Instruction Fetch and Decode Techniques [**JS**]	
	9/4	Fri	Recitation 2: Ensure Team Formations & Discussion of Modern Superscalar Out-of-Order Processors	
Week 3	9/8	Tues	Lecture 5: Register Renaming & Out-of-Order Execution [**JS**]	
	9/10	Thur	Lecture 6: Dynamic Instruction Scheduling Techniques [**JS**]	
	9/11	Fri	Recitation 3: Lab 2 Preview & Tomasulo's Algorithm	
	Lab 2: CPU Design: Superscalar Out-of-Order Structures / Cache Coherence			Lab 2
Week 4	9/15	Tues	Lecture 7: Dealing with Long Latency Memory Instructions [**OH**]	
	9/17	Thur	Lecture 8: Modern Memory Hierarchy & Data Storage [**OH**]	
	9/18	Fri	Recitation 4: Cache Coherence, Memory Consistency, Asym. Cores	
Week 5	9/22	Tues	Lecture 9: System Level Memory Prefetching Techniques [**TH/LC**]	
	9/24	Thur	Lecture 10: Operating System & Runtime Environment [**SM/LC**]	
	9/25	Fri	Recitation 5: Operating Systems & Memory Subsystem	
Week 6	9/29	Tues	Lecture 11: Parallel Programming & Threading Models [**OH**]	
	10/1	Thur	Lecture 12: Modern Heterogeneous CPU-GPU Architectures [**OH**]	
	10/2	Fri	Recitation 6: Parallel Programming & Heterogeneous Architectures	
Week 7	**10/6**	**Tues**	**MIDTERM EXAM #1**	
	10/8	Thur	Lecture 13: GPU and TPU for Accelerating AI Workloads [**OH**]	
	10/9	Fri	Recitation 7: Unified Memory & Lab 3 Logistics (Async)	
	10/12-10/16		**NO CLASSES** - Fall Break!!!	
	Lab 3: Heterogeneous Cores: Profiling Snapdragon SoC			Lab 3
Week 8	10/20	Tues	Lecture 14: Survey of Cloud & Edge AI Inference Accelerators [**OH**]	
	10/22	Thur	Lecture 15: Research on Sparse Tensor Compilation and Mapping for Heterogeneous Domain-Specific Accelerators [**OH**]	
	10/23	Fri	Recitation 8: Exam 1 Post Mortem & Dataflow	
Week 9	10/27	Tues	Lecture 16: Performance, Power, Energy Efficiency Iron Laws [**JS**]	
	10/29	Thur	Lecture 17: (Re)Emergence of VLIW in NPU and LPU Designs [**JS**]	
	10/30	Fri	Recitation 9: VLIW & Power/Energy Efficiency	
Week 10	11/3	Tues	**NO CLASS** - Democracy Day	
	11/5	Thur	Lecture 18: Return of the CPU? Overview of NVIDIA Vera CPU [**JS**]	
	11/6	Fri	Recitation 10: CPU Re-emergence / Lab Assistance	
Week 11	11/10	Tues	Lecture 19: Experimental Characterization of LLM Inference Workloads on Heterogeneous CPU-GPU Systems [**JS**]	
	11/12	Thur	Lecture 20: Neuromorphic & NeuroAI Computing Systems [**JS/LC**]	
	11/13	Fri	Recitation 11: Neuromorphic Computing	
	Lab 4: Heterogeneous Cores: Heterogeneous Systems and Algorithms			Lab 4
Week 12	**11/17**	**Tues**	**MIDTERM EXAM #2**	
	11/19	Thur	Lecture 21: Special Guest #1 Career Story of a Computer Architect [**Bryan Black** - Moto, Intel, AMD, **Chipletz**] CMU alum	
	11/20	Fri	Recitation 12: Lab Assistance & Exam #2 Post Mortem	
Week 13	11/24	Tues	**NO CLASS**	
	11/26	Thur	**NO CLASS** - Thanksgiving Holiday	
	11/27	Fri	**NO CLASS** - Thanksgiving Holiday	
Week 14	12/1	Tues	Lecture 22: Special Guest #2 Career Story of a Computer Architect [**Trung Diep** - Intel, Rambus, **Marvell**] CMU alum	
	12/3	Thur	Lecture 23: Special Guest #3 Career Story of a Computer Architect [**Bob Rychlik** - Intel, **Qualcomm**] CMU alum	
	12/4	Fri	Recitation 13: Lab Assistance & Course Wrapup	
	12/9	Wed	Lab 4 Team Project Submission Deadline (no late submission)	

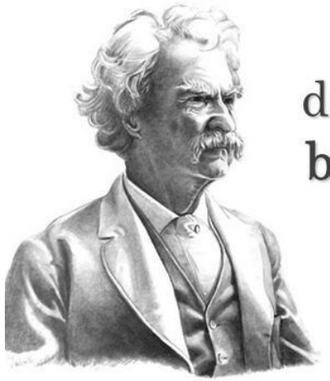
18-740: Modern Computer Architecture & Design

Lecture 1:
"Computer Architecture & Processor Design"

A. **Big Picture of Computing Landscape**

B. **Instruction Set Architecture (ISA)**

C. **Historical Perspective on Computing**

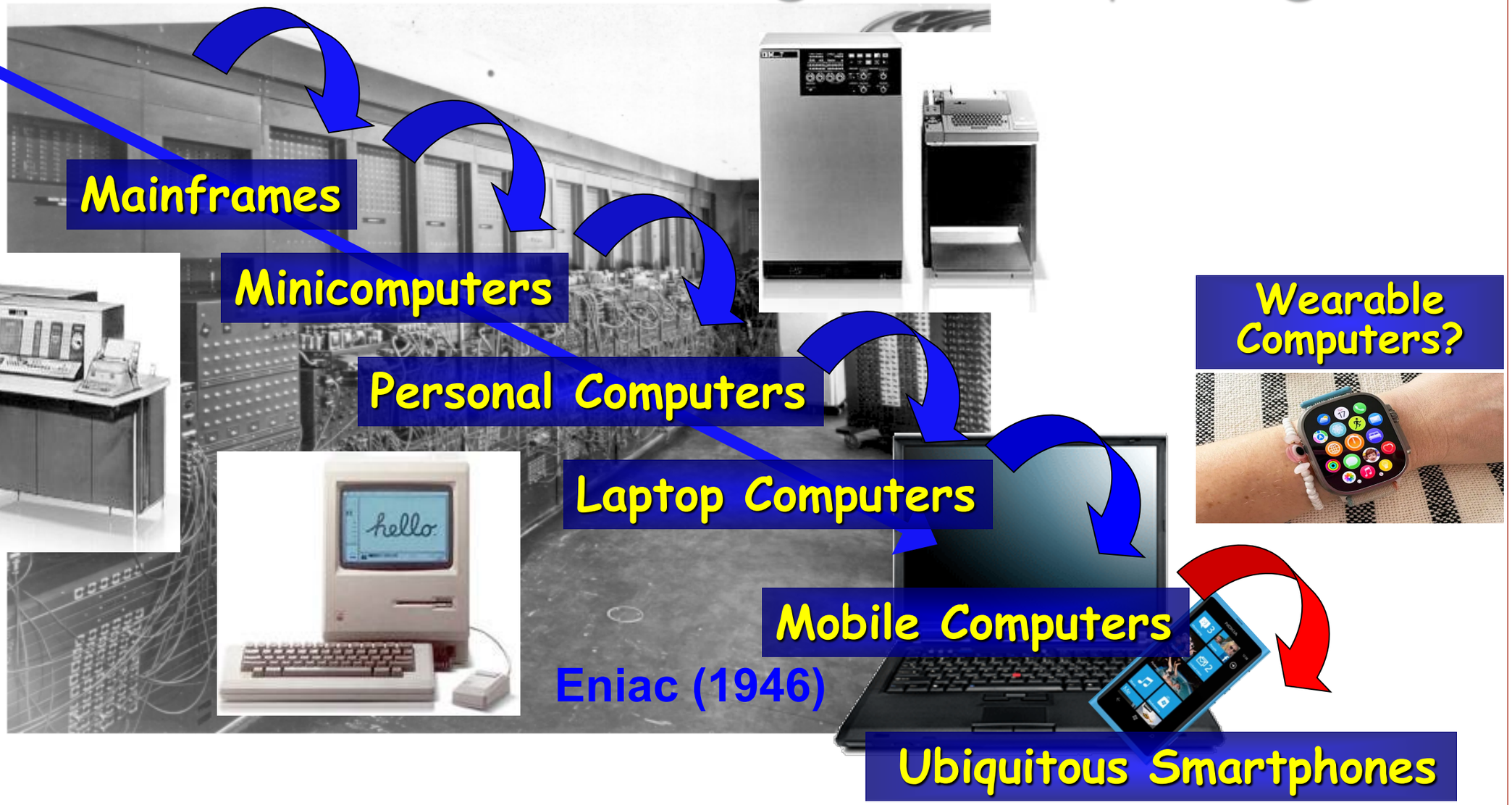


History
doesn't repeat itself,
but it does rhyme.

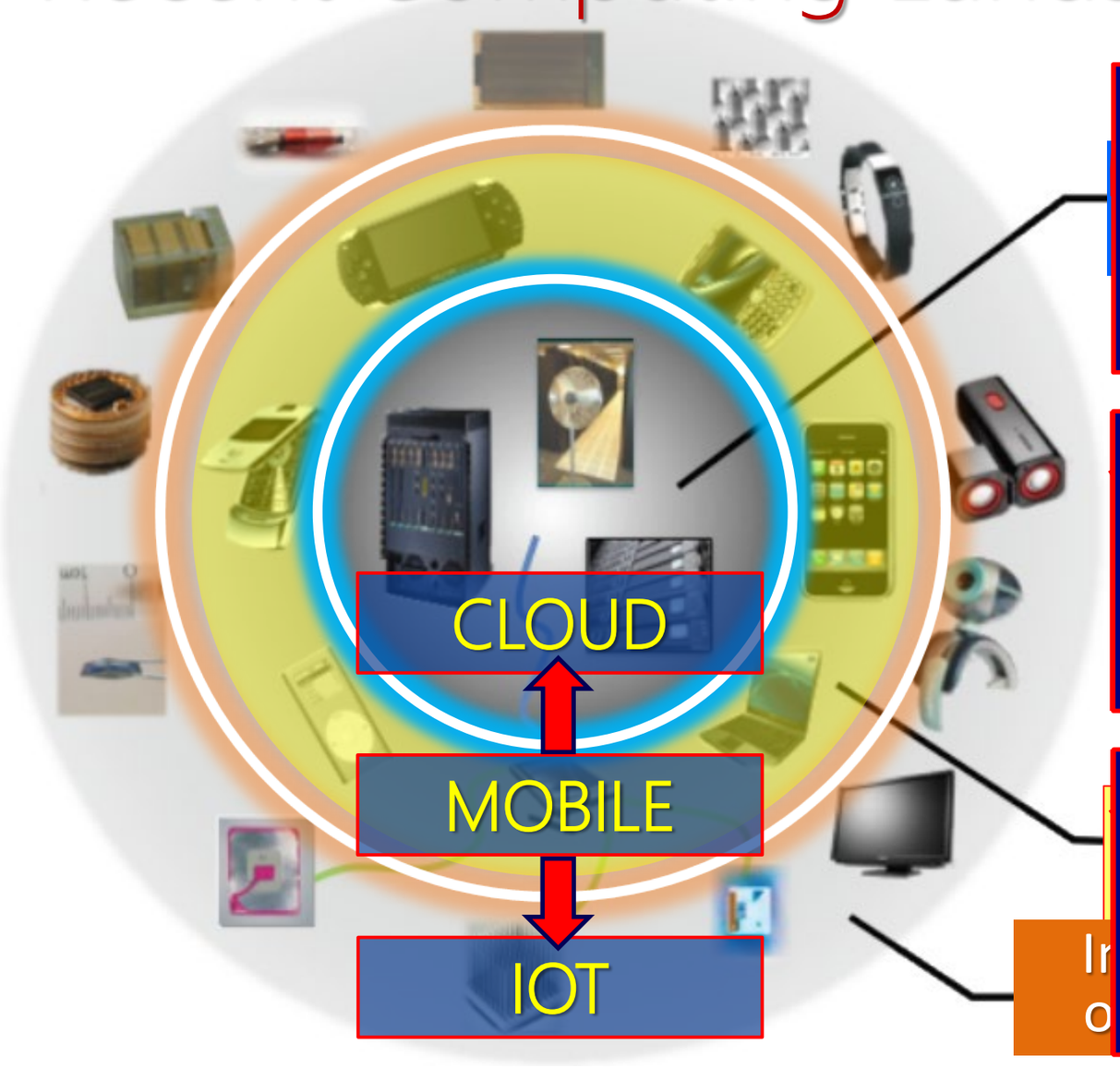
Mark Twain

Electrical & Computer ENGINEERING

The 80 Years of "Modern" Digital Computing . . .



Recent Computing Landscape… (2010-2020)



Shifting Cloud Computing to the Edge:

Push cloud infrastructure to wireless 5G edge
Alleviate core network congestion/saturation
Reduce services delivery latency to end users

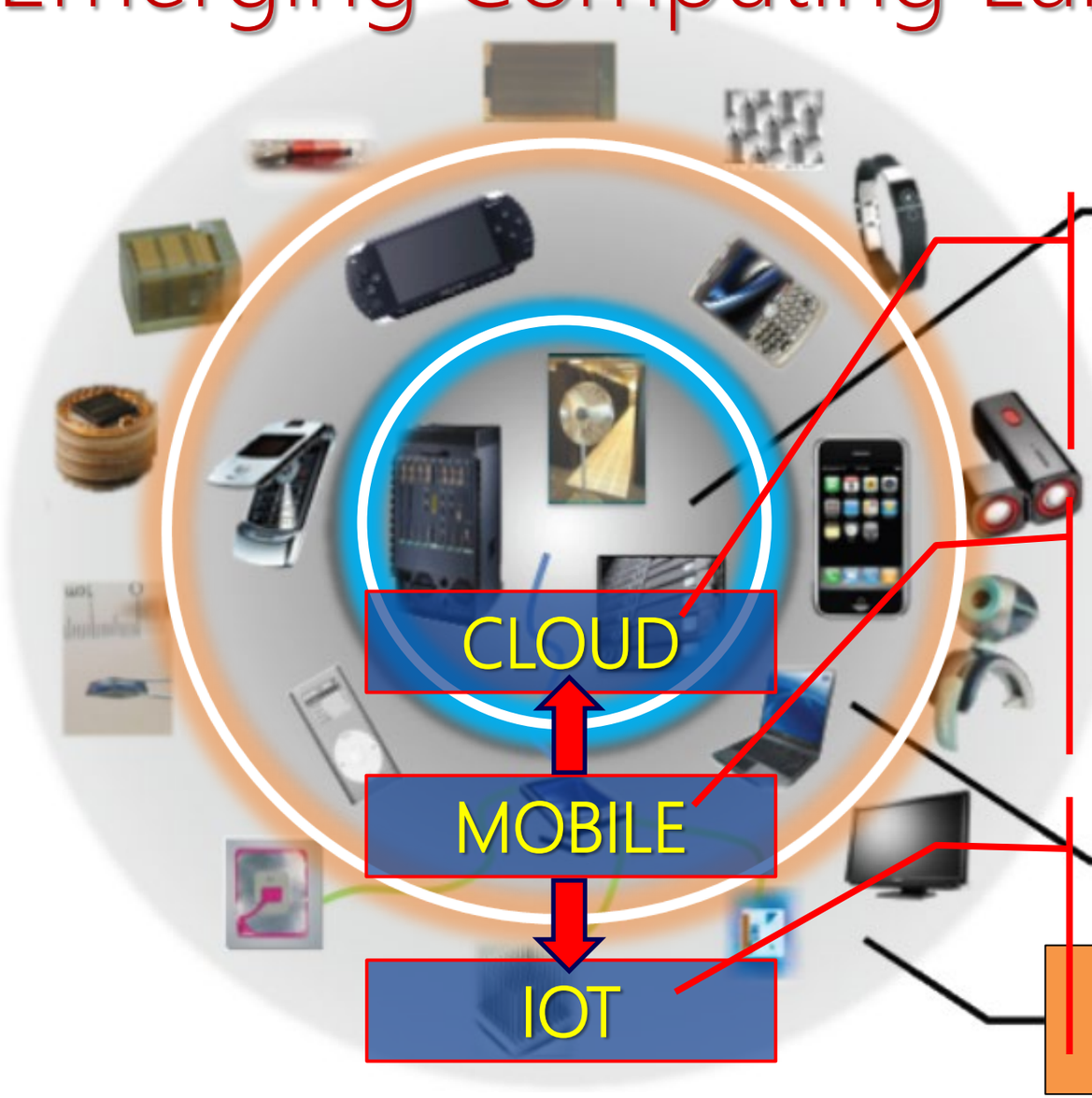
New Seamless Mobile User Experience:

Re-factoring mobile devices form factors
One mobile "persona" across many devices
Mobile & edge devices as hub of human mobility

Global Scale Sensing for Societal Good:

Deliver real IoT values to mobile population
Moving from Eulerian to Lagrangian Sensing
Distributed edge sensing, analytics, & services

Emerging Computing Landscape… (2020-2030)



Cloud Comp…

Dominance of LLM & Gen. **AI** Workloads:

Highly Competitive Cloud Infrastructures & Services
Pursuing Vertical Integration with HW to Differentiate

Off-Load **AI** Compute to Edge & Mobile:

Convergence of Computing and Communication Infra
Necessity of 1000x Improvement on Energy Efficiency

Global Distributed **AI** Sensory Network:

Necessity of Online, On-Device, Continuous Learning
Disruptive AI Inversion by IoT Sensory Signal Processing

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Engineering Design: The Builder's Perspective



SPECIFICATION: Behavioral description of *"What does it do?"*

Synthesis: Search for possible solutions; pick best one. *Creative process*

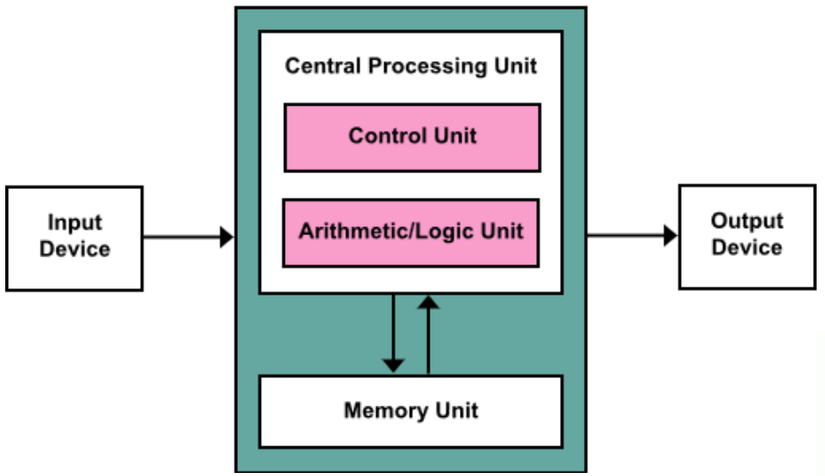
IMPLEMENTATION: Structural description of *"How is it constructed?"*

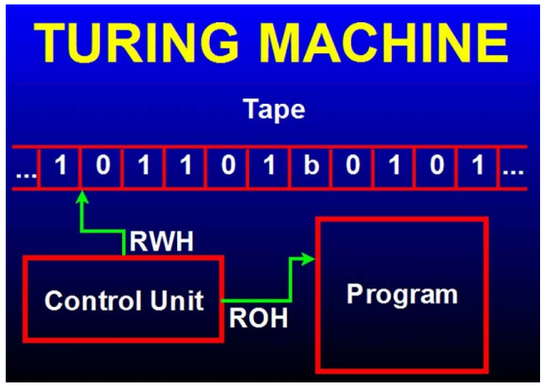
Analysis: Validate if the design meets the specification.

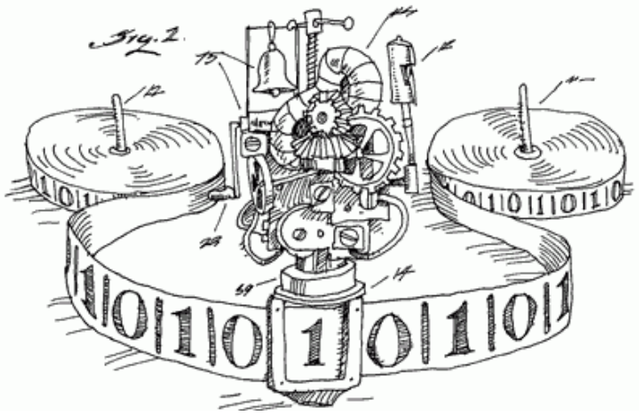
"Does it do the right thing?" + "How well does it perform?"

The Von Neumann Stored Program Computer

- **The Classic Von Neumann Computation Model**: Proposed in 1945 by John Von Neumann and others (Alan **Turing**, J. Presper **Eckert**, and John **Mauchly**).
- **A "Stored Program Computer"**
  1. **One CPU**
     - One Control Unit
       - Program Counter
       - Instruction Register
     - One ALU (Data Path)
  2. **Monolithic Memory**
     - Data Store
     - Instruction Store
  3. **Sequential Execution Semantics**
     - **Instructions from an Instruction Set (ISA)**



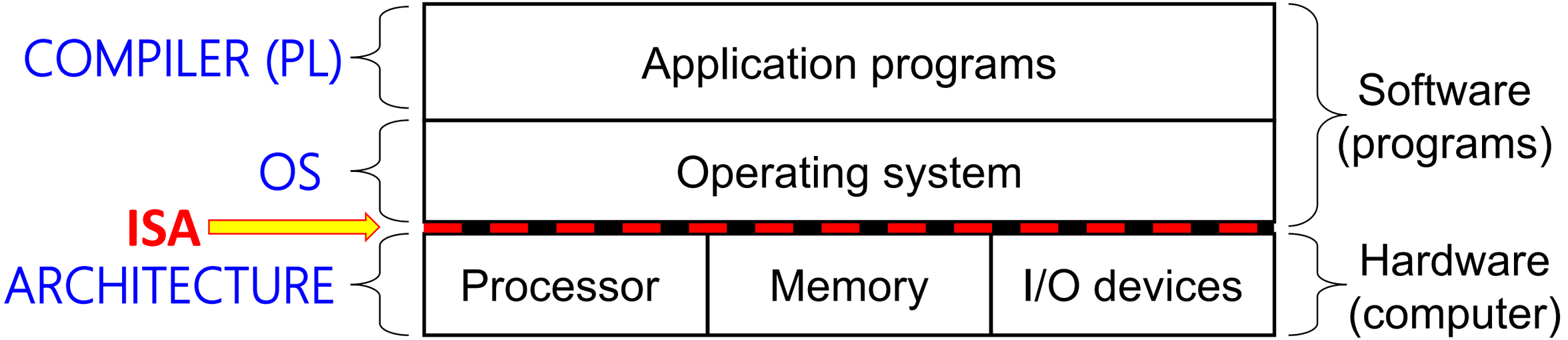






Anatomy of a Computer System: (ISA)

- What is a Computer System?
  - = [Software + Hardware] = [Programs + Computer]
  - = [Application program + OS] + [Processor + Memory + I/O Devices]
  - Programming Languages + Operating Systems + Computer Architecture



Computer Hardware/Software Interface (HSI)

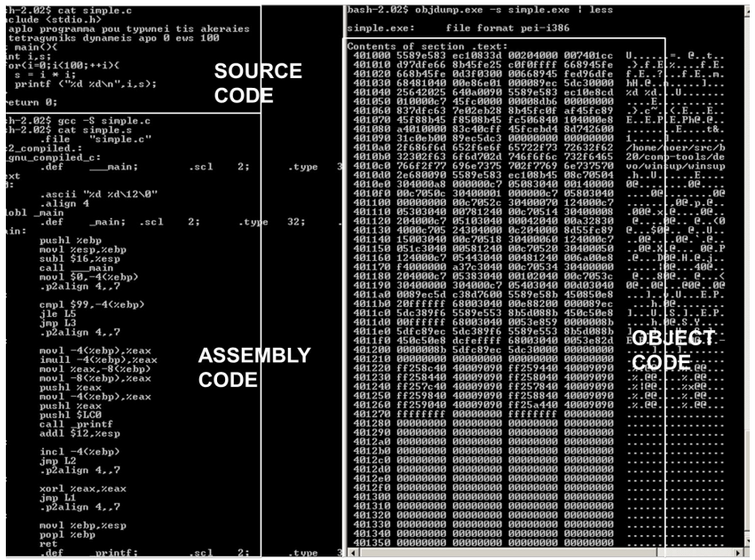
Software Engineering

Program Development

Application Software
- Program development
- Program compilation

SPECIFICATION

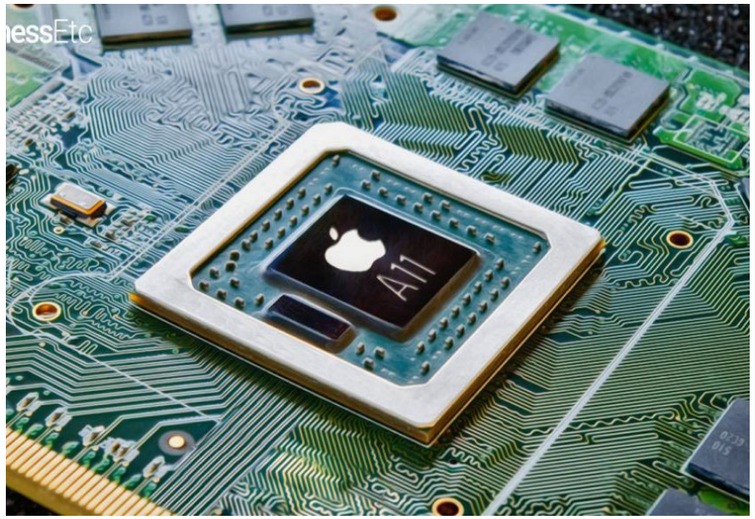
Instruction Set Architecture (ISA)



Computer Engineering

Processor Design

Hardware Technology
- Program execution
- Program performance



IMPLEMENTATION

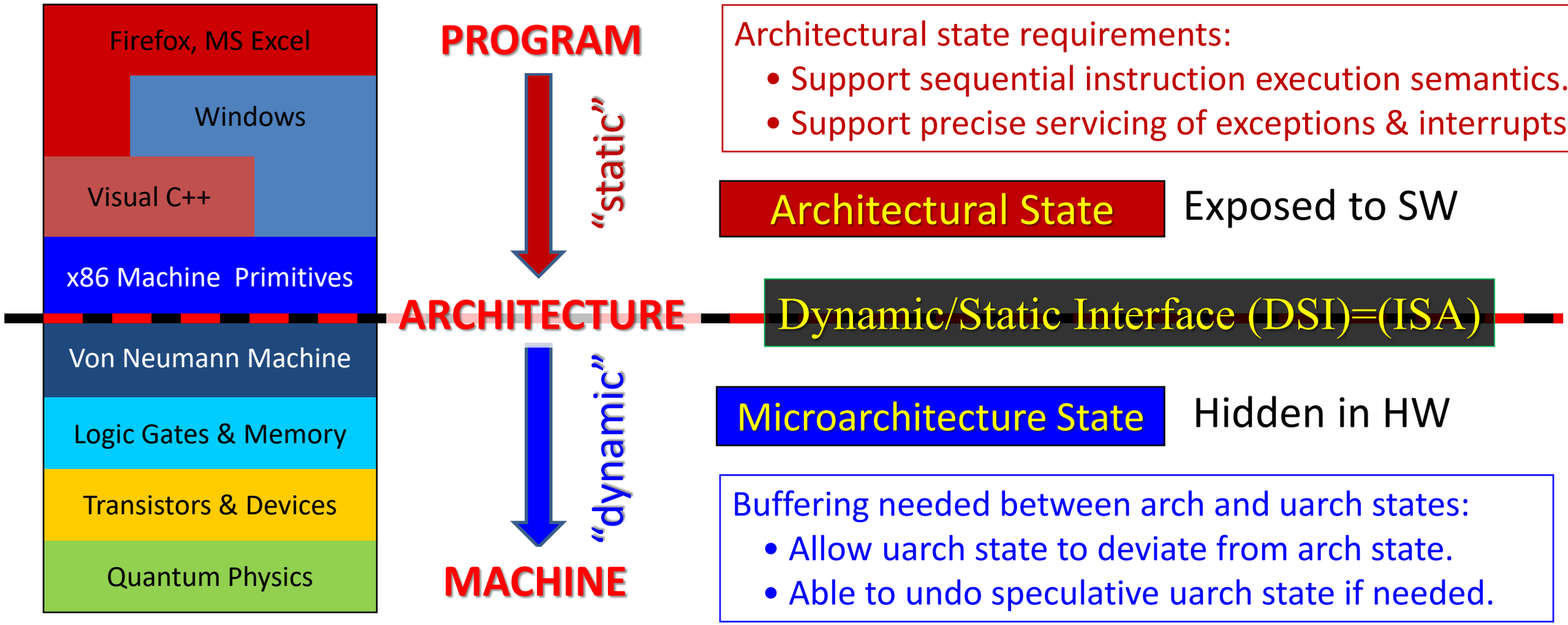
The Processor Design Process for an Architect

Architecture (ISA):
- Processor State
- Instruction Set
  - Instruction Formats
  - Register Transfers



Microarchitecture:
- HW Organization
  - Sequential Buffers
  - Combination Logic
- RTL Implementations

Computer Dynamic/Static Interface (DSI)



Architectural state requirements:
- Support sequential instruction execution semantics.
- Support precise servicing of exceptions & interrupts.

Architectural State — Exposed to SW

Dynamic/Static Interface (DSI)=(ISA)

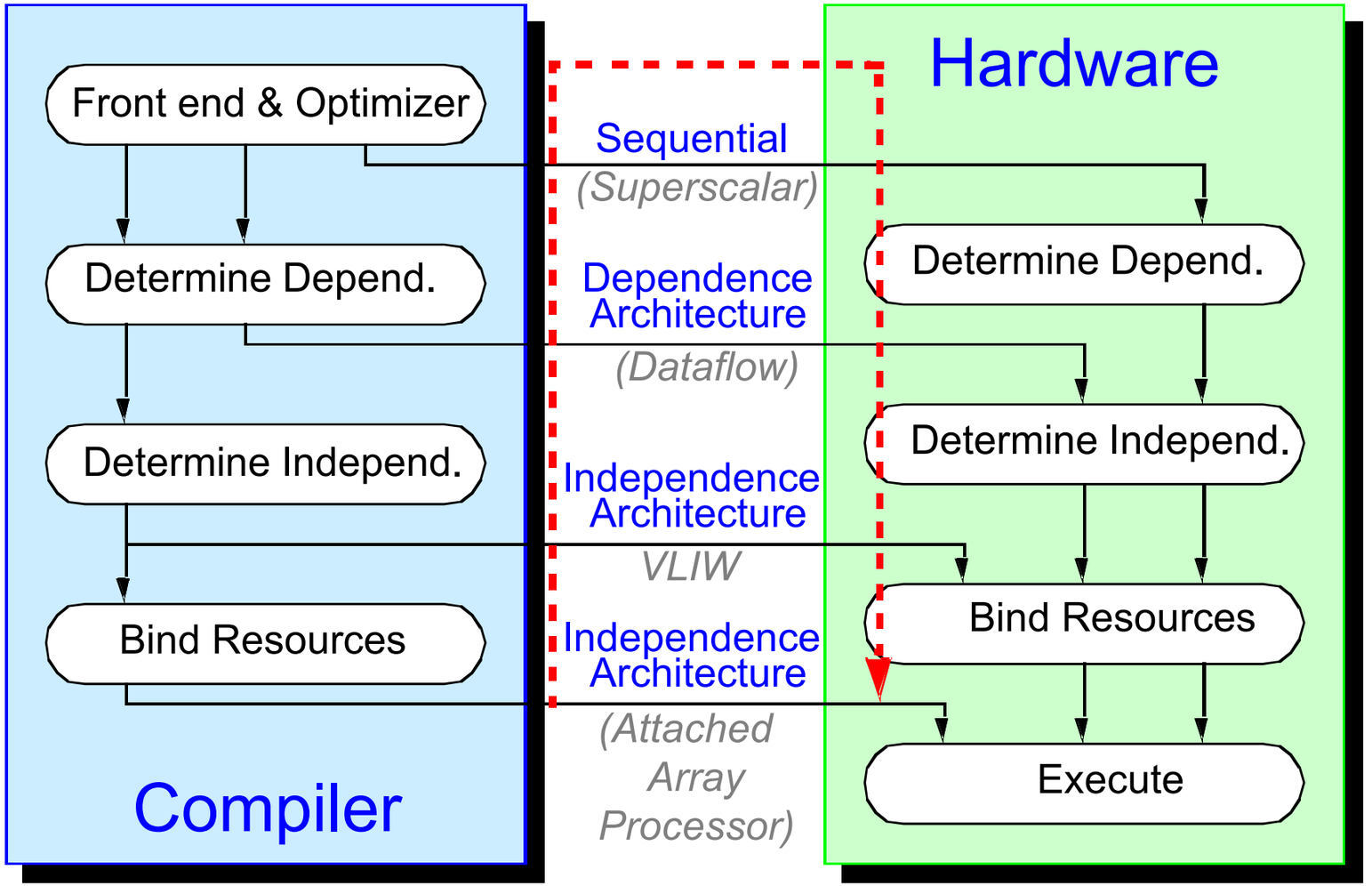
Microarchitecture State — Hidden in HW

Buffering needed between arch and uarch states:
- Allow uarch state to deviate from arch state.
- Able to undo speculative uarch state if needed.

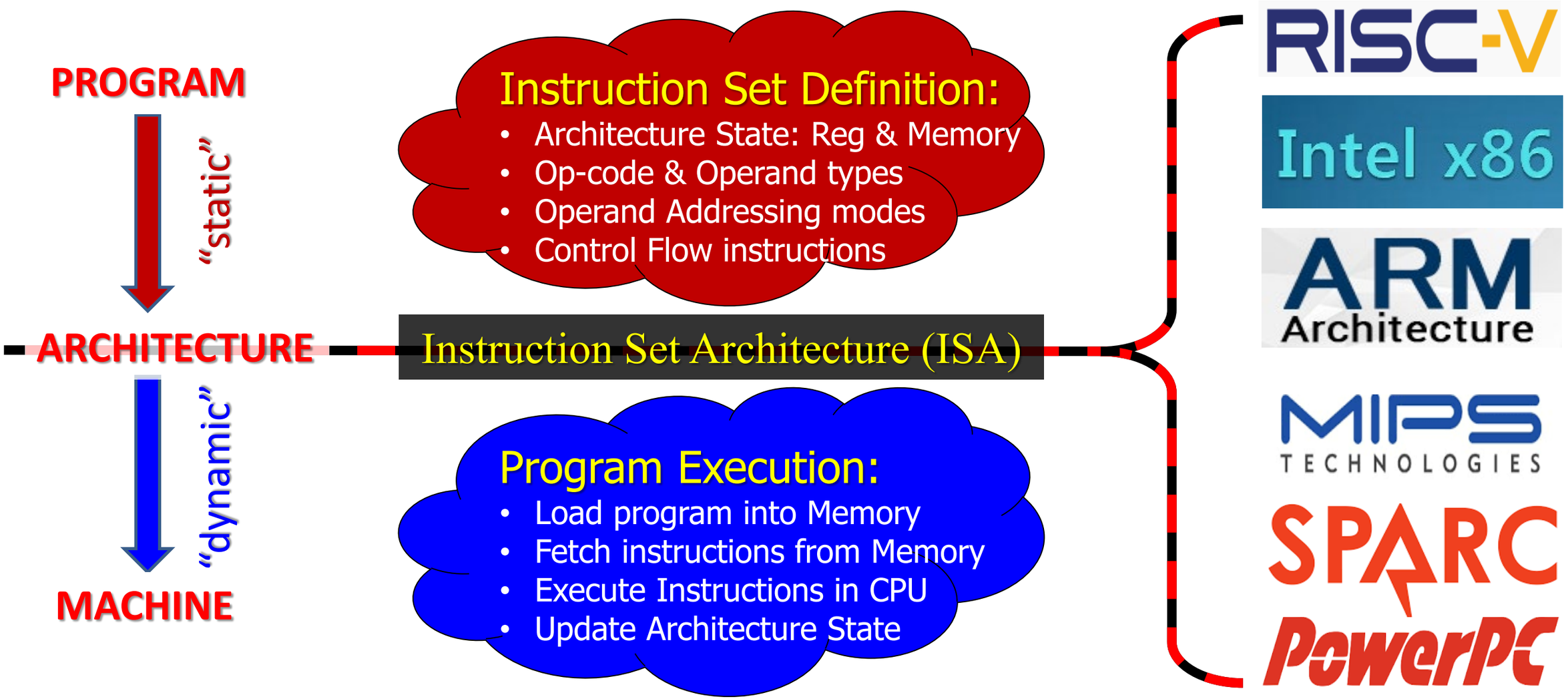
DSI = ISA = a contract between the program and the machine.

[B. Rau & J. Fisher, 1993]

Dynamic/Static Design Space: DSI Placement



Commercially Successful ISAs



18-740: Modern Computer Architecture & Design

Lecture 1:
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A. Big Picture of Computing Landscape

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"Iron Law" of Processor Performance

$$1/\text{ComputerPerformance} = \frac{\text{Time}}{\text{Program}}$$

$$= \frac{\text{Instructions}}{\text{Program}} \times \frac{\text{Cycles}}{\text{Instruction}} \times \frac{\text{Time}}{\text{Cycle}}$$

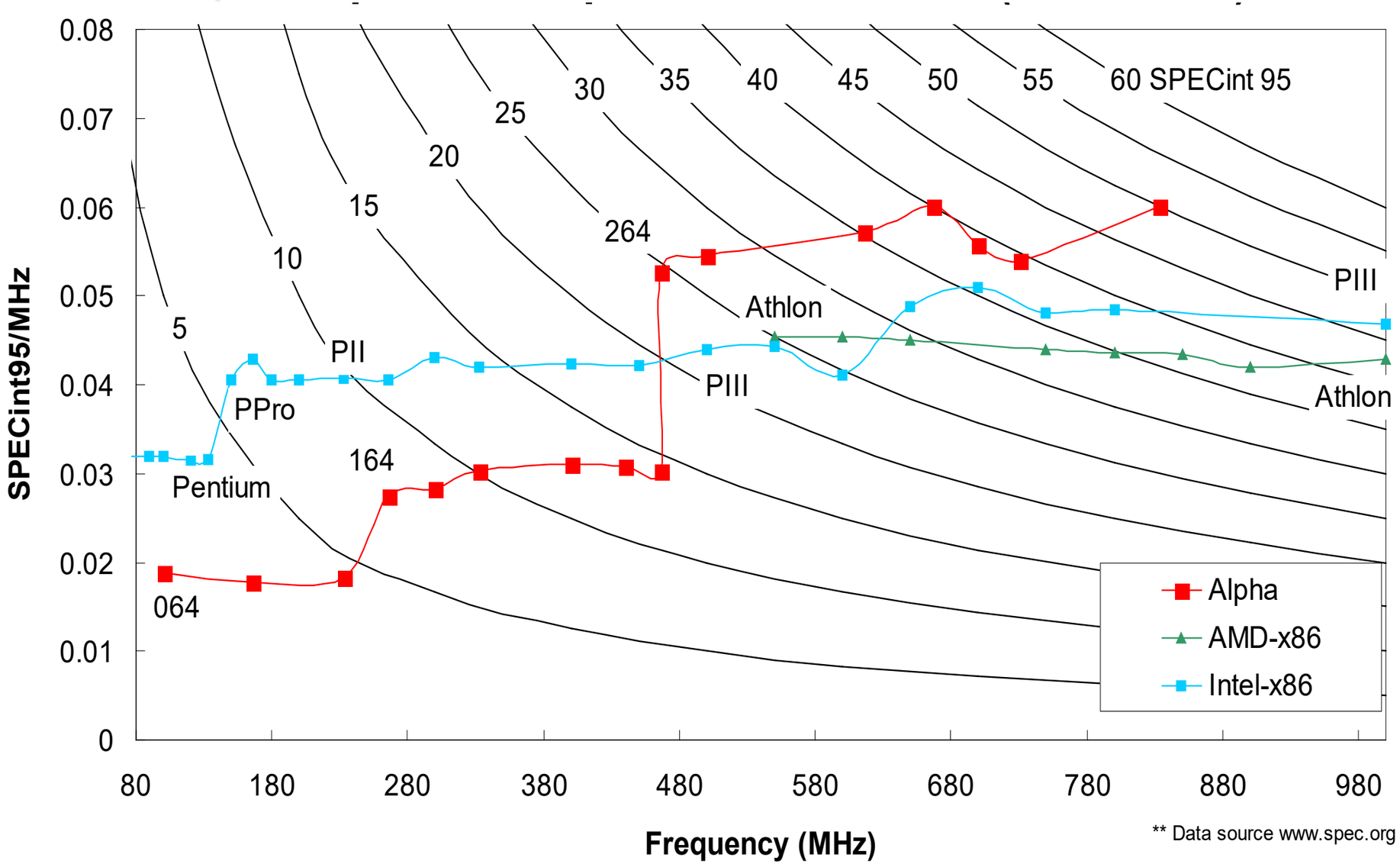
(inst. count) (CPI) (cycle time)

Architecture → Implementation → Realization

Compiler Designer Processor Designer Chip Designer

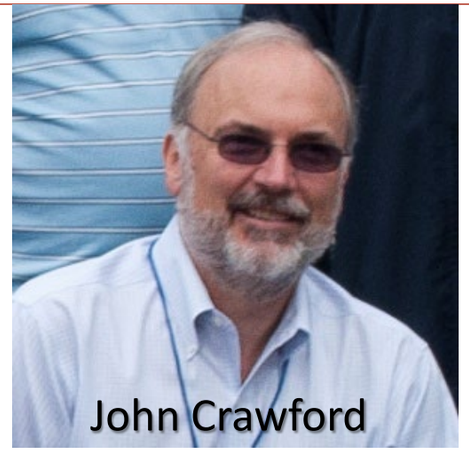
- In the 1980's: **(pipelining)**
  - CPI: 5.0 → 1.2
- In the 1990's: **(superscalar)**
  - CPI: 1.2 → 0.5
  - IPC: 0.8 → 2.0
- In the 2000's: **(multicores)**
  - Hit power wall !!!
  - ILP → TLP

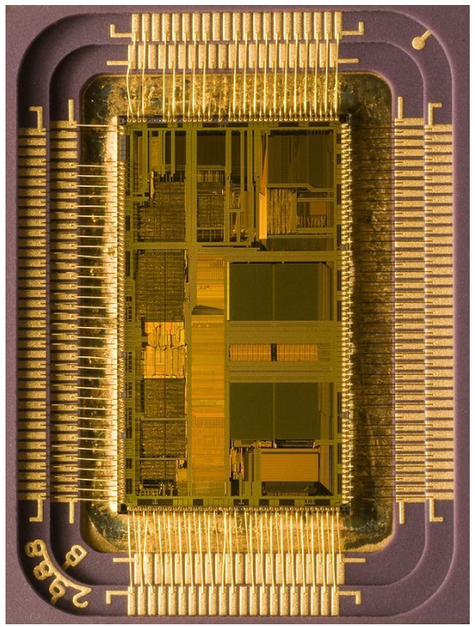
Landscape of Processor Families [SPECint95]

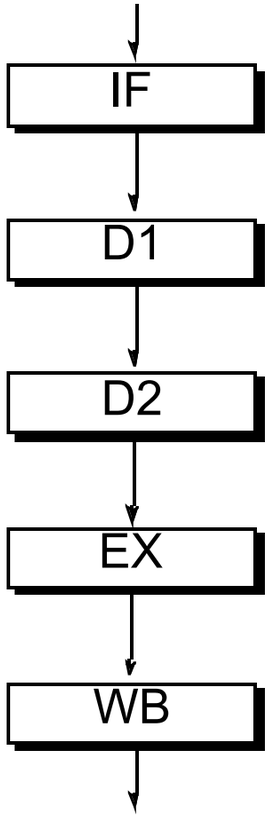


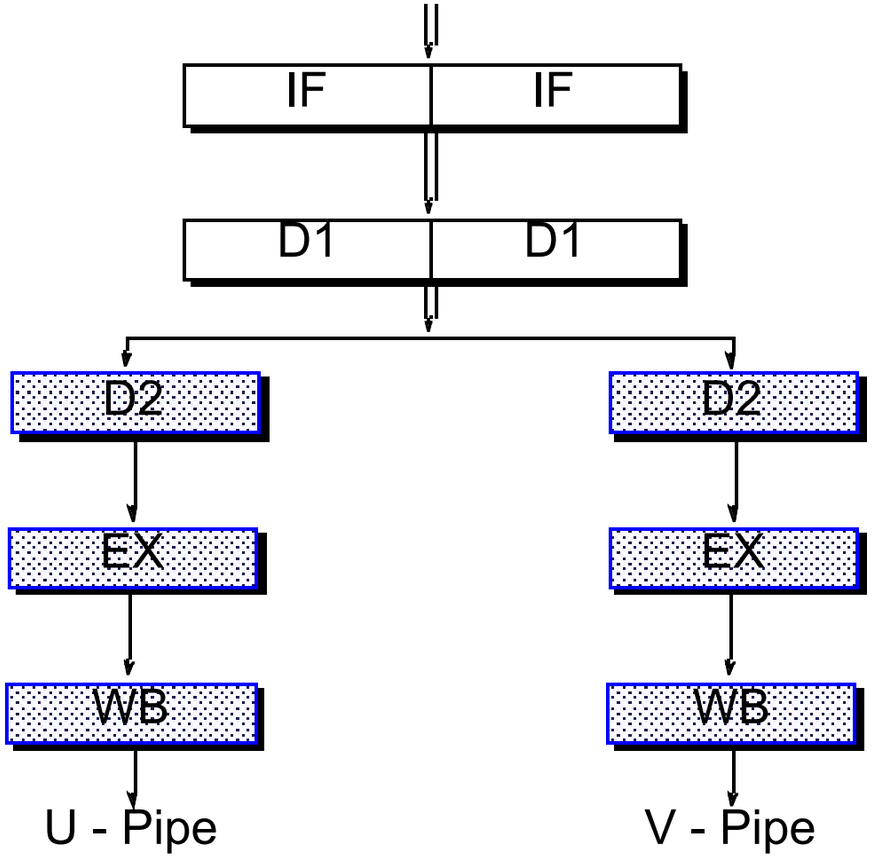
** Data source www.spec.org

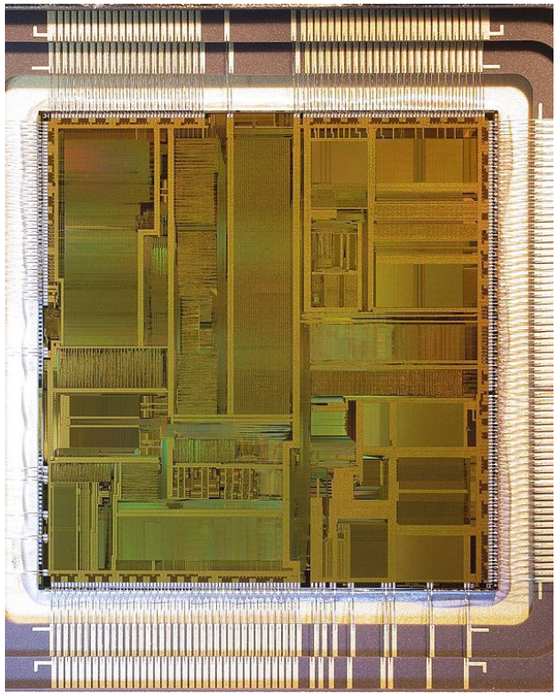
Intel 80486 & Pentium (P5) Processors







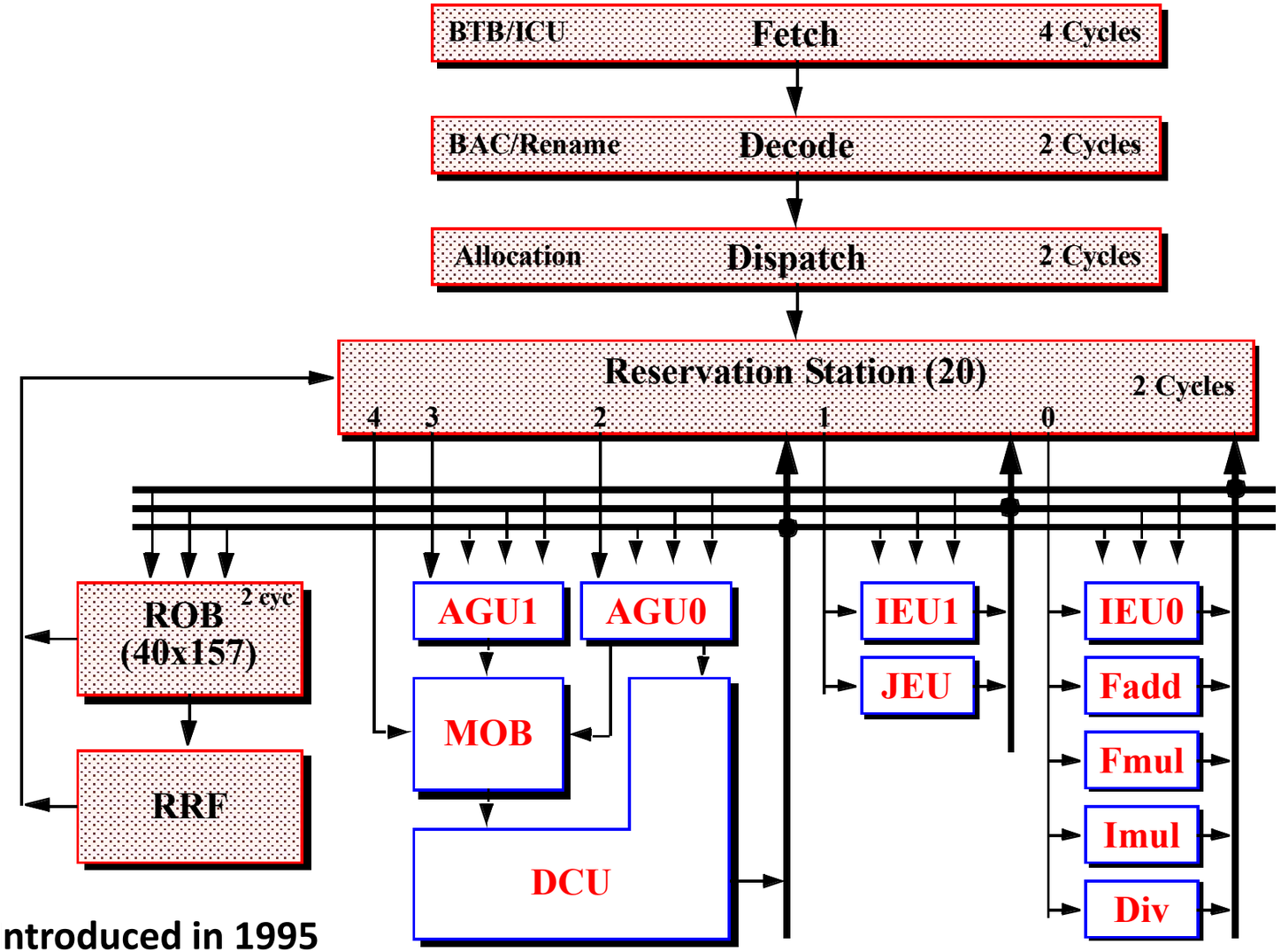




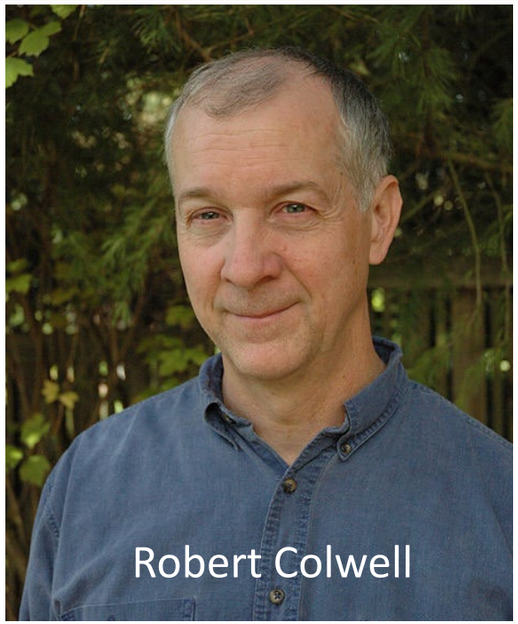
Introduced in 1989

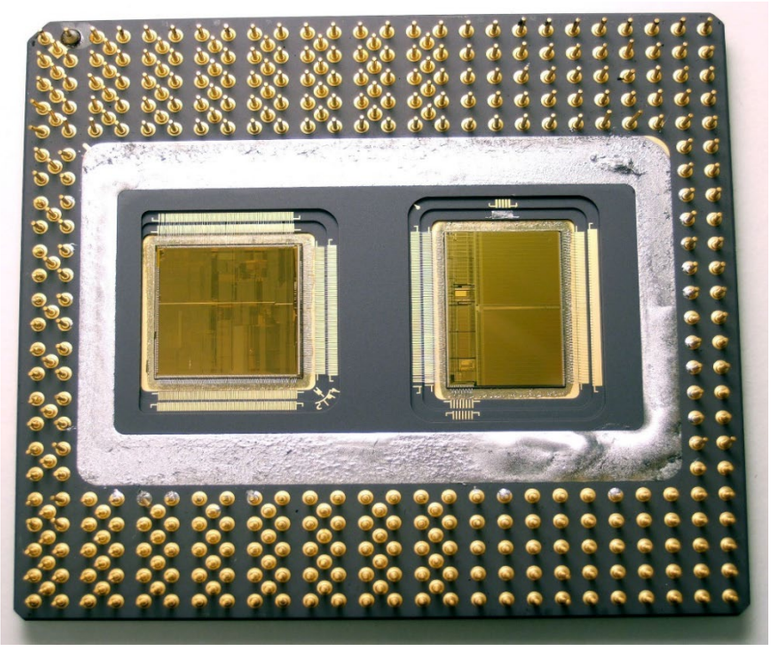
Introduced in 1993

Intel Pentium Pro (P6) O3 Processor



Introduced in 1995

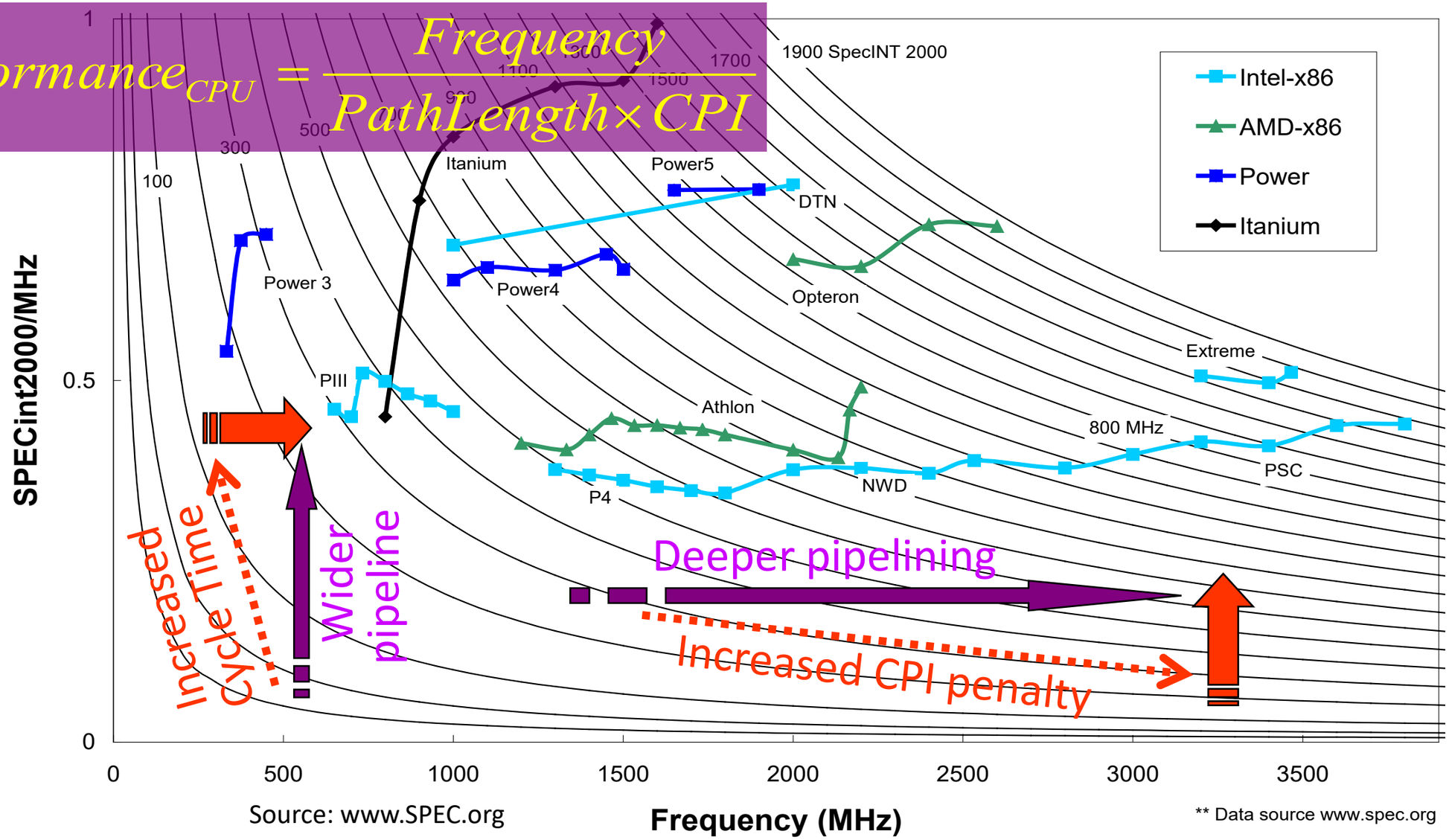




[John DeVale & Bryan Black, 2005]

Landscape of Processor Families [SPECint2000]

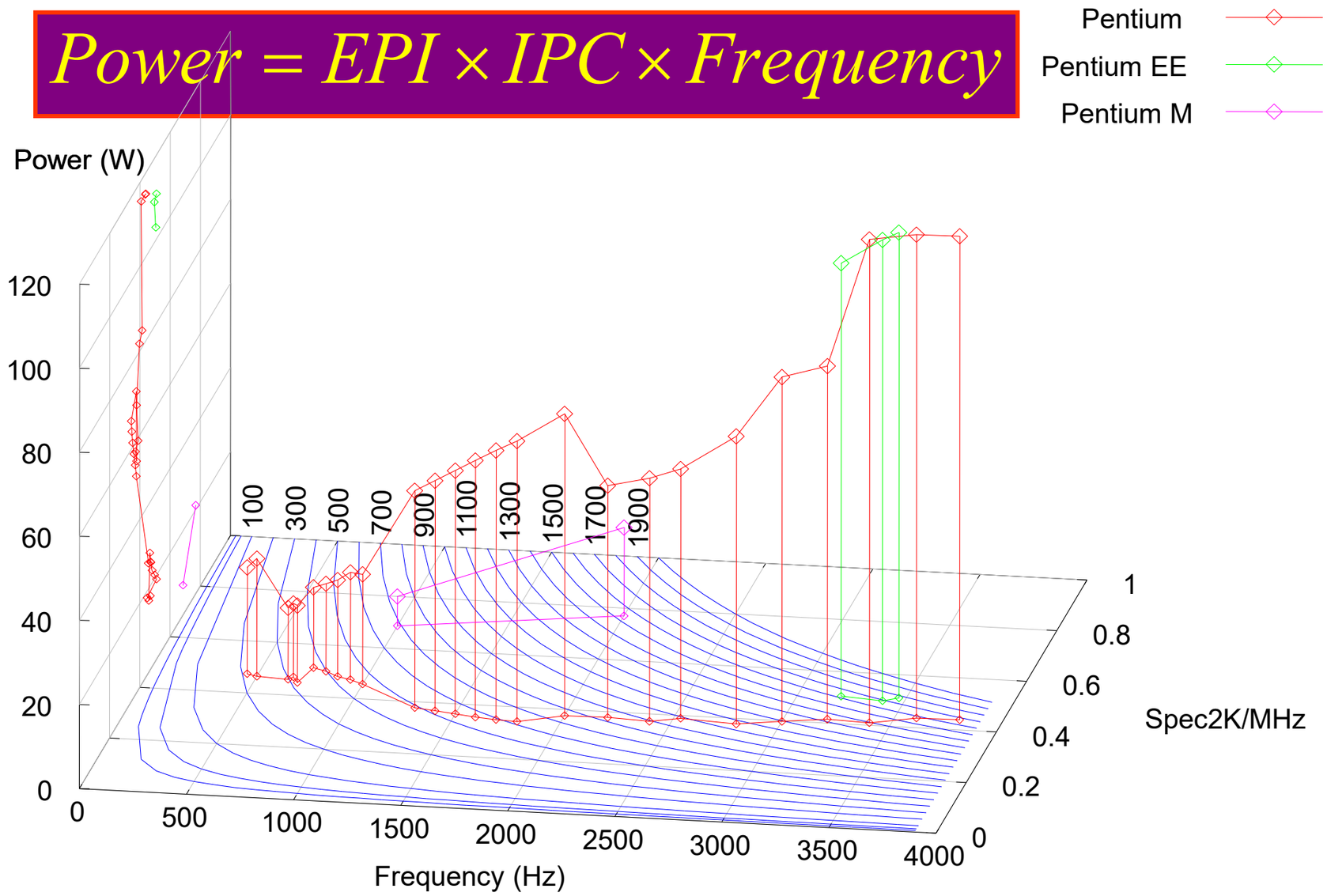
$$Performance_{CPU} = \frac{Frequency}{PathLength \times CPI}$$



[John DeVale & Bryan Black, 2005]

Iron Law of Performance and Power

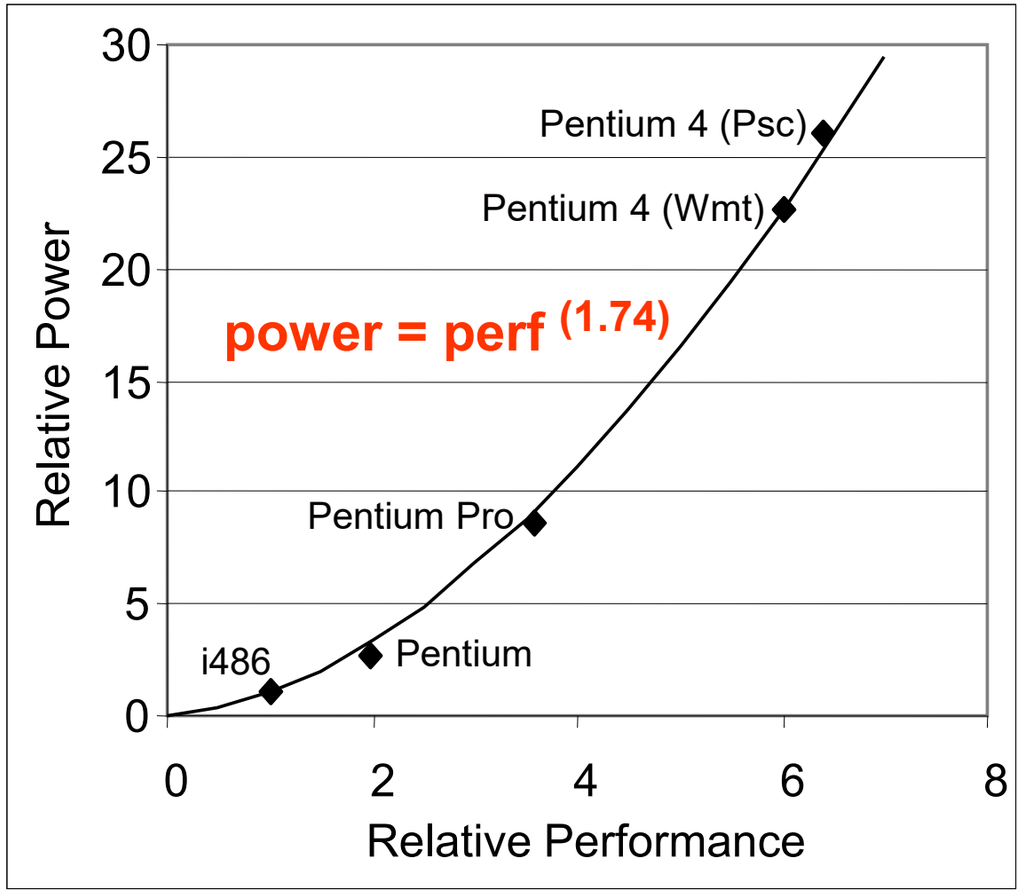
$$Power = EPI \times IPC \times Frequency$$



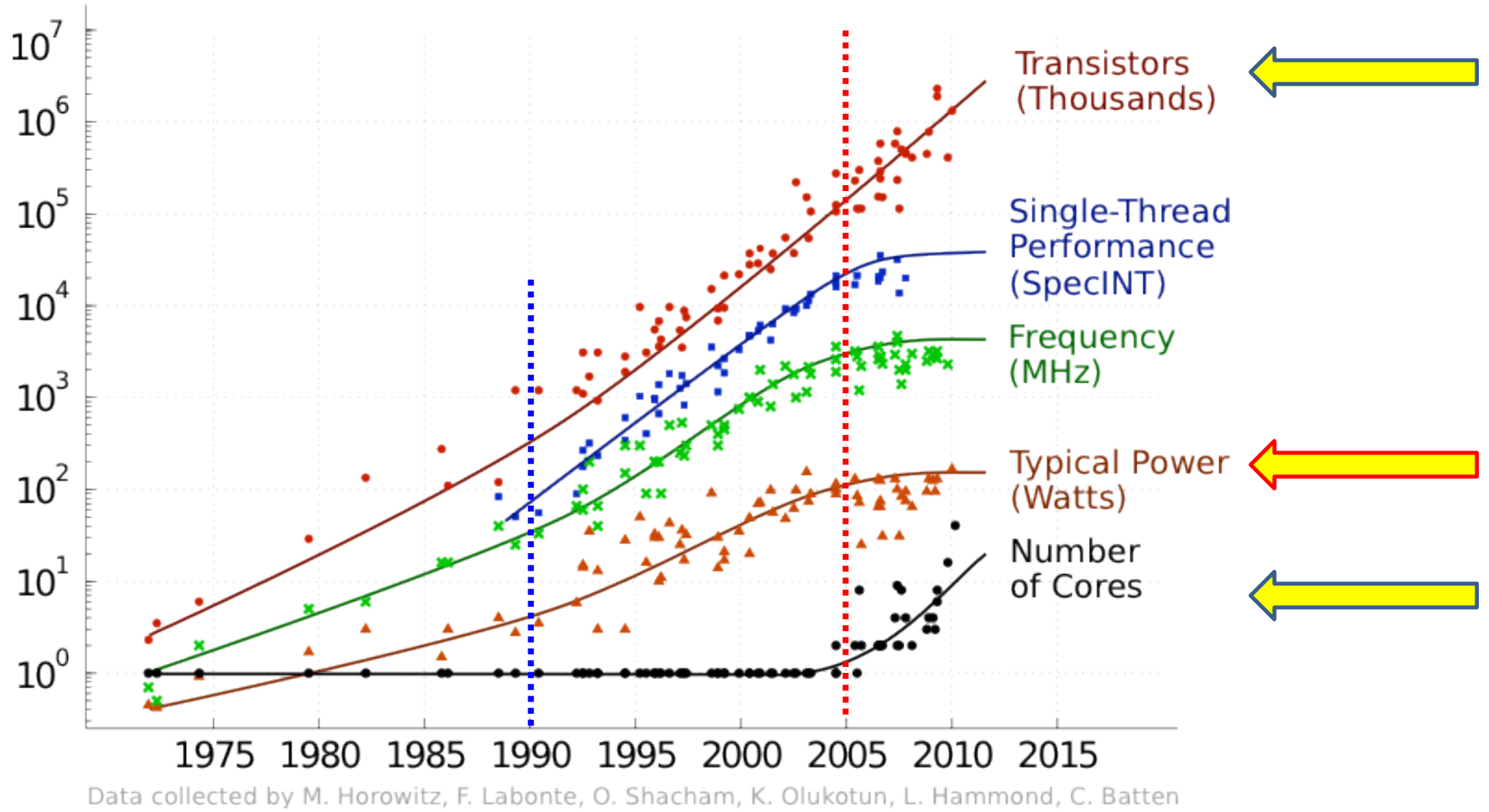
[Ed Grochowski, Intel, 2004]

Power and Latency (Single-Thread) Performance

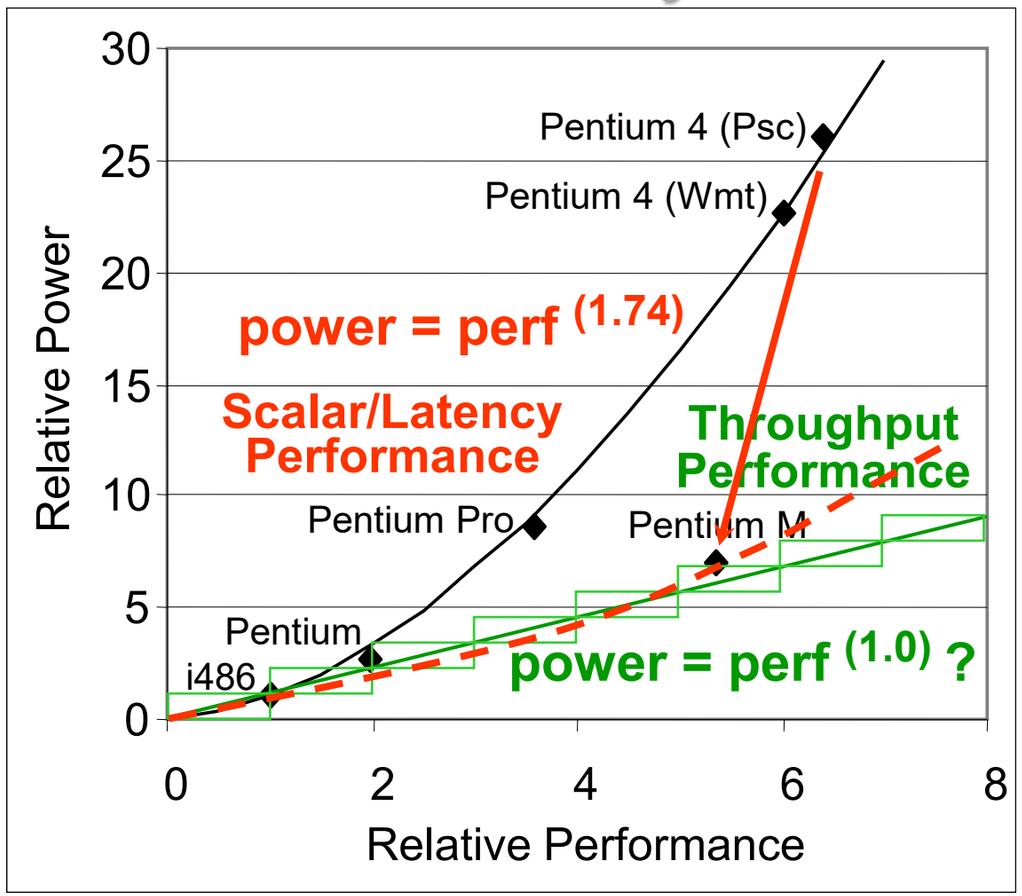
- For comparison
  - Factor out contributions due to process technology
  - Keep contributions due to microarchitecture design
  - Normalize to i486™ processor
- Relative to i486™ Pentium® 4 (Wmt) processor is
  - 6x faster (2X IPC at 3X frequency)
  - 23x higher power
  - Spending 4 units of power for every 1 unit of scalar performance



Processor Scaling Since ~AD 2005



Power and Latency/Throughput Performance



[Ed Grochowski, 2005]

CPU	EPI
I486	7 nj
P5	10 nj
P6	17 nj
P4P-wmt	27 nj
P4P-psc	29 nj
Pentium M	9 nj

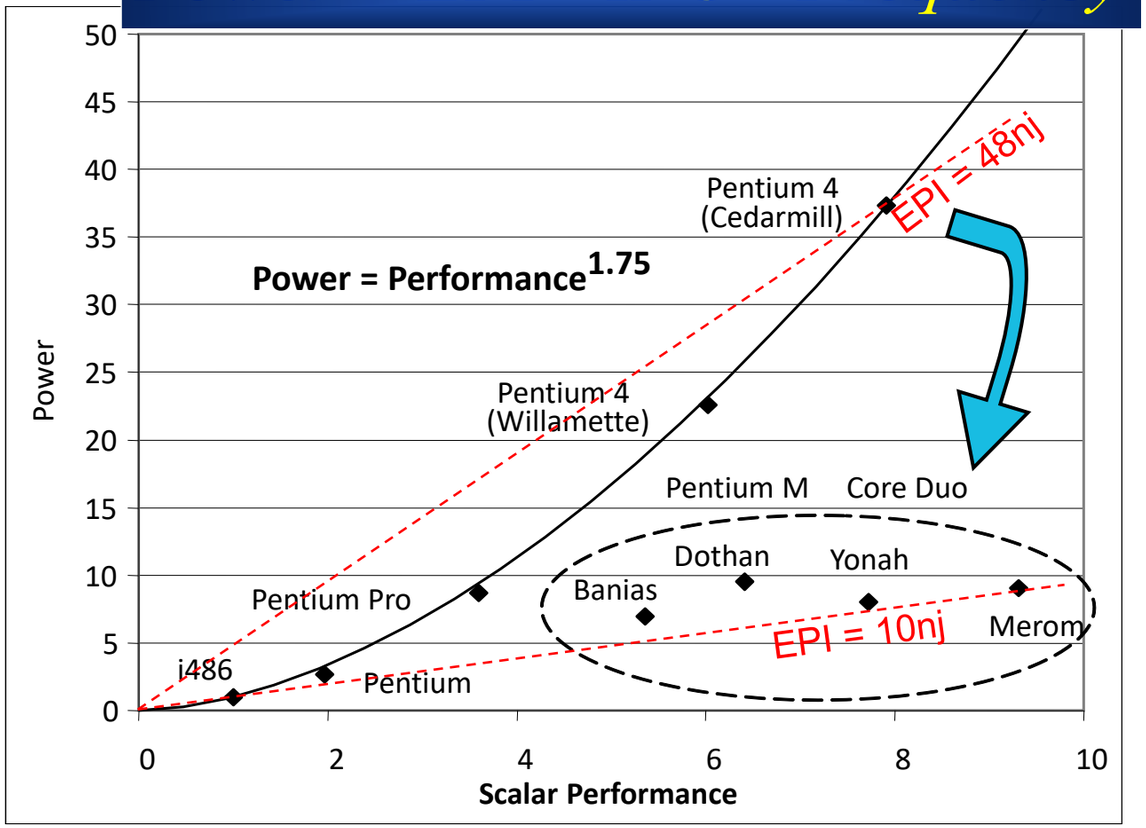
Low EPI

- Assume a large-scale MCP with potentially many cores
- Replication of cores results in (nearly) proportional increases to both throughput performance and power (hopefully).

Power/Performance (EPI) Evolution

$$Power = EPI \times IPC \times Frequency$$

[Ed Grochowski, 2004]



Intel Microprocessors	EPI (nj) 65nm at 1.33v
i486	10
Pentium	14
Pentium Pro	24
Pentium 4 (WMT)	38
Pentium 4 (CDM)	48
Pentium M (Banias)	13
Pentium M (Dothan)	15
Core Duo (Yonah)	11
Core Duo (Merom)	10

- Power: single core power (relative to i486 baseline)
- Performance: SPECint performance (relative to i486 baseline)
- EPI: average energy spent per instruction (in nano-joules)

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Evolution of Von Neumann Architecture Implementations:

- ~~SP: Sequential Processors (direct implementation of sequential execution)~~
- PP: Pipelined Processors (overlapped execution of in-order instructions)
- SSP: Superscalar Processors (out-of-order execution of multiple instructions)
- MCP: Multi-core Processors (concurrent multi-threads and/or multi-programs)
- HMP: Heterogeneous Multi-Processors (concurrent multi-tasks on diverse co-processors)

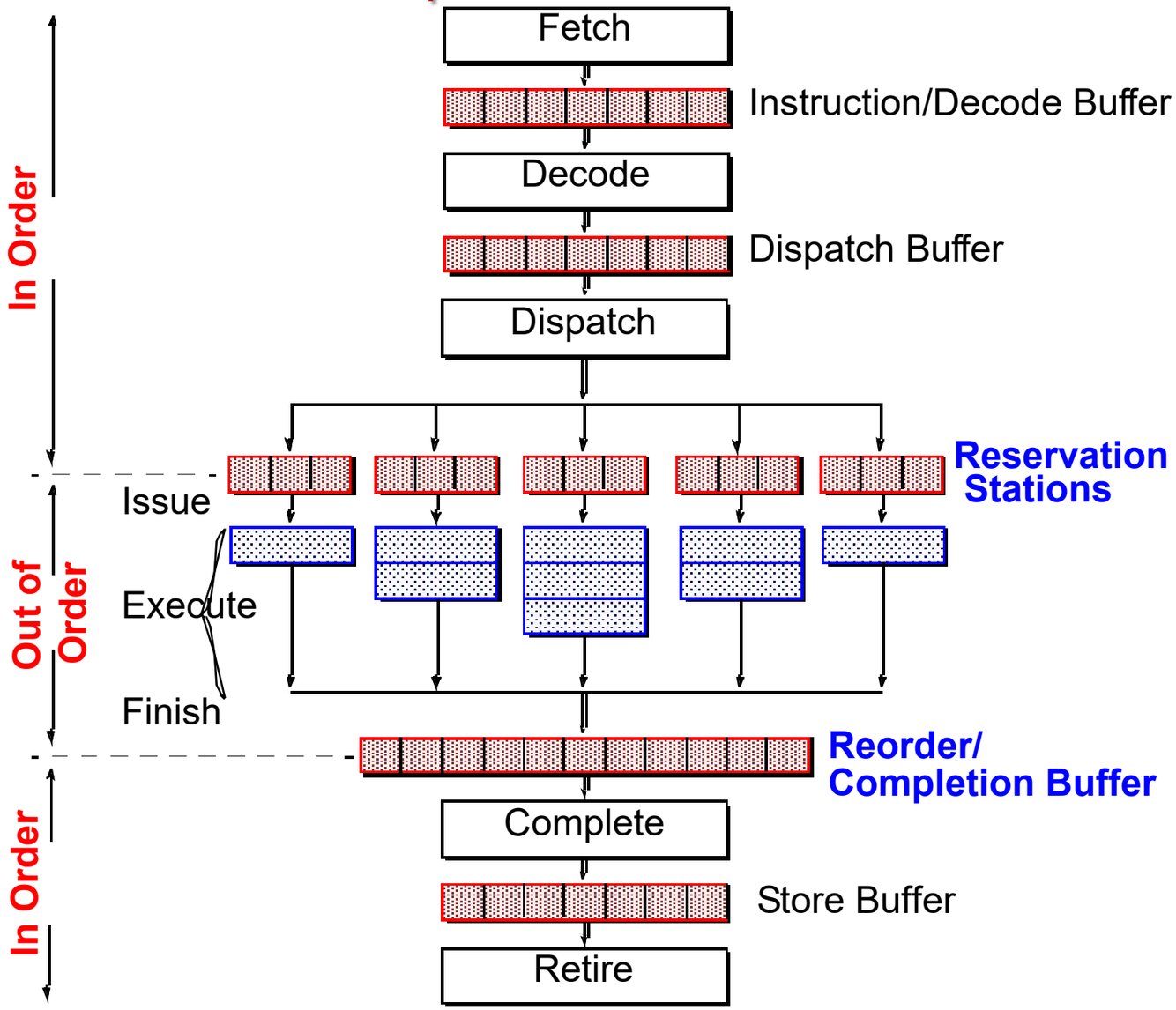
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Parallelisms for **Performance** (→ for **Power** Reduction → for **Energy** Efficiency)

- **ILP**: Basic Block (exploit ILP in PP, SSP)
- **ILP**: Loop Iteration (exploit ILP in SSP, VLIW)
- **DLP**: Data Set (exploit DLP in VP/SIMD, GPU)
- **TLP**: Thread/Task (exploit TLP in MCP, HMP)
- **SLP**: System/Workload (exploit SLP in Multi-Computer Clusters)

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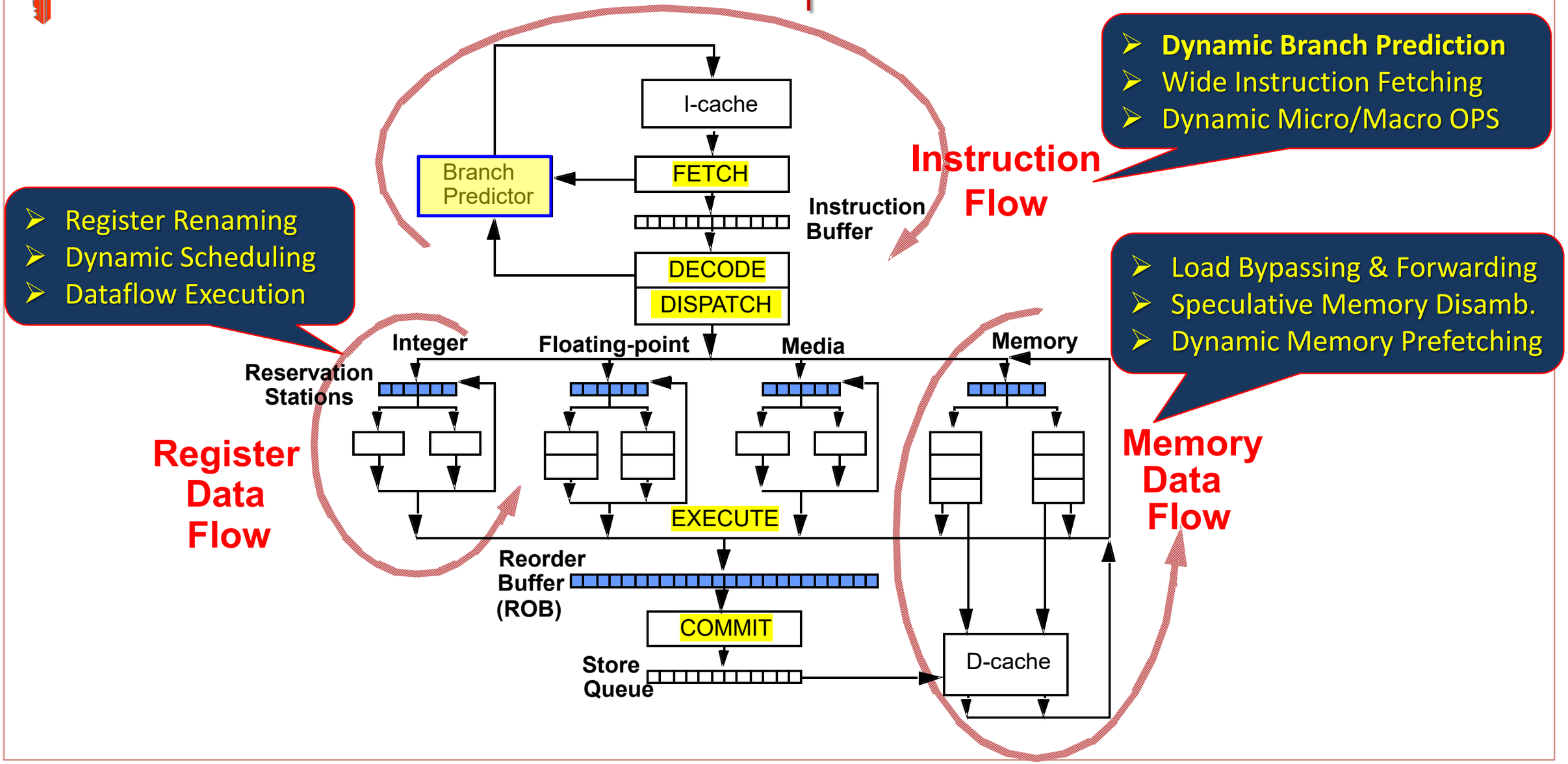
Modern Superscalar Processor Organization



- Buffers provide decoupling
- In OOO designs they also facilitate (re-)ordering
- More details on specific buffers to follow

[Mike Johnson, *Superscalar Microprocessor Design*, Prentice-Hall, 1991]

Three Flow Paths of Superscalar Processors



https://www.amazon.com/Superscalar-Microprocessors-Design-Mike-Johnson/dp/0138756341

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Lecture 2: "Modern Superscalar Out-of-Order Processors"

John Paul Shen
August 27, 2026

Next Time ...

- **Lab 0** posted: start working on it; bring your questions to recitation
  - Lab 0 tasks are due on Sunday, August 30, 2026 (11:59pm ET)
  - Lab 0 questionnaire due Monday, August 31, 2026 (11:59pm ET)
- Look for team partners for the team lab assignments (**Labs 1-4**)
- **Required Reading Assignment**:
  - **Read Chapters 2 and 4 of Shen and Lipasti (SnL)**

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Questions?

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